



Uncovering the Effect of Model Weights on Politically Based Prompts



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Introduction

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Large language models (LLMs) are increasingly used to generate politically relevant content. However, their outputs can vary widely depending on architecture, training data, and model weights. Understanding how weight adjustments affect politically based prompts is essential for evaluating potential bias, fairness, and reliability.

Abstract

This project investigates how variations in large language model (LLM) weights affect the outcomes of politically based prompts. Using LLama-3.1/4, we evaluate how weight adjustments influence generated text, bias, and consistency in responses. The research provides insights into model transparency and the implications of fine-tuning for politically sensitive contexts.

Limitations

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- Limited to English-language prompts
- Experiments focused on U.S.-based political discourse
- Analysis restricted to LLama-3.1 and LLama-4 architectures

Methods/Materials

Methods

- Frameworks: PyTorch, TensorFlow, Hugging Face Transformers
- Models: LLama-3.1 and LLama-4
- Approach:
 - Conducted research using open weights to test political prompts and analyze responses
 - Applied activation engineering techniques to modify cluster behaviors and observe outcomes
 - Leveraged PyTorch, TensorFlow, and Transformers to carry out experiments
 - Performed fine-tuning and weight modification
 - Comparative analysis of bias, coherence, and factual alignment across models

Results and Conclusions

Results

- Uncovered systematic response patterns in LLMs when exposed to politically sensitive prompts
- Weight adjustments significantly altered political framing and sentiment in responses
- LLama-4 showed greater consistency but amplified certain biases more strongly than LLama-3.1
- Identified scenarios where minimal weight changes had disproportionate effects on outputs

Conclusions

- Model weights play a critical role in shaping political content generated by LLMs
- Bias mitigation requires systematic weight-level evaluation rather than surface-level prompt filtering
- Findings provide key insights for the lab’s ongoing evaluation of model bias and alignment

Future Research

Future ideas for research

- Expand testing to multilingual political prompts
- Compare LLama models with GPT and Mistral frameworks
- Develop systematic weight-mapping techniques for bias auditing

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